

## Calculus II 第十章泰勒级数的应用(求极限,求和函数和泰勒展开式)

### 考点一:用泰勒级数求极限.

- 1.(2022年期末) Use Taylor series to evaluate  $\lim_{n \rightarrow \infty} (n^3 \sin \frac{2}{n} - 2n^2)$ .
- 2.(2021年期中) Find  $\lim_{n \rightarrow \infty} ((n^2 - n)e^{\frac{1}{n}} - \sqrt{n^4 + 1})$ .
- 3.(2020年期末) Find the real numbers  $a, b (b \neq 0)$ , which satisfy  $\lim_{x \rightarrow 0} \frac{\cos(\sin x) - \sqrt{1-x^2}}{x^a} = b$ .
4. Using Taylor series to compute the following series.
  - (1)  $\lim_{x \rightarrow 1} \frac{x^2 - 1}{\ln x}$
  - (2)  $\lim_{x \rightarrow \infty} (\sqrt[6]{x^6 + x^5} - \sqrt[6]{x^6 - x^5})$
  - (3)  $\lim_{x \rightarrow \infty} x^{\frac{3}{2}} (\sqrt{x+1} + \sqrt{x-1} - 2\sqrt{x})$
  - (4)  $\lim_{x \rightarrow \infty} [x - x^2 \ln(1 + \frac{1}{x})]$

### 考点二:求函数的泰勒展开式.

- 1.(2023年期末) Using known Taylor series expansions, write the Taylor series for the function  $f(x) = \frac{4}{x^2 - 2x + 5} + \ln x$  at  $x = 1$  in the interval  $(0, 2)$ .
- 1.(2021年期末) Find the Maclaurin series for  $f(x) = \int_0^{x^2} \frac{1}{1-t} dt$ ,  $-1 < x < 1$ .
2. (2019年期中) Find the Taylor series for  $f(x) = \ln(x + \sqrt{x^2 + 1})$  at  $x = 0$ .
3. (2018年期中) Find the Maclaurin series for the function  $f(x) = \frac{1}{(2-x)^2}$ .
4. Find the Maclaurin series for the function  $f(x) = \tan^{-1}(\frac{1+x}{1-x})$ .

### 考点三:求泰勒级数的和函数.

- 1.(2021年期中) For the power series  $f(x) = \sum_{n=1}^{\infty} \frac{n+2}{n(n+1)} x^n$ ,
  - (1) For what values of  $x$  does the power series converge?
  - (2) Find the sum of the series within the interval of convergence.
2. Find the sum function for the following series  $\sum_{n=0}^{\infty} \frac{n^2+1}{2^n n!} x^n$ .